



Tracing the Milky Way warp with red clump kinematics

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ABSTRACT

Galactic warping is almost ubiquitous amongst observed external disc galaxies, and the Milky Way is itself no exception, exhibiting a strong warp in both its stellar and gaseous components. The main limitation to the further study of this warp is uncertain distance measurements as a consequence of the obscuring dust in the Galactic mid-plane. This work utilises photometry from the AllWISE mid-infrared survey along with *Gaia* DR3 to trace the kinematics of the warp in red clump stars to $R = 16$ kpc. Jeans’ first equation is used to estimate vertical velocities within a rigidly precessing disc with two models being tested. A fixed model held the line of nodes (ϕ'_w) and precession rate (ω_p) constant across the Galactic disc, while a twisted model allowed both of these parameters to vary linearly with Galactocentric radius. It was found that the twisted model best fit the data, forming a trailing spiral line of nodes starting at $\phi'_w \approx 98.5^\circ$ at 10 kpc and varying by $8.5^\circ \text{ kpc}^{-1}$, whereas the fixed model resulted in a constant line of nodes $\phi'_w \approx 125.4^\circ$. Both models found prograde precession rates for the warp reaching approximately $-12 \text{ km s}^{-1} \text{ kpc}^{-1}$ at $R = 16$ kpc, with the twisted model yielding a precession rate gradient of $-1.9 \text{ km s}^{-1} \text{ kpc}^{-2}$ from $R = 10 - 16$ kpc increasing in the prograde direction. The maximum warp amplitudes for the models varied, with the fixed model estimating steep warping reaching 4.1 kpc at $R = 16$ kpc, compared to the twisted model’s 1.1 kpc, which plateaus beyond $R \approx 14$ kpc. These results indicate that the warp in the Milky Way is not fixed but twisting with Galactocentric radius and precessing, suggesting that it is a complex and evolving feature of the disc.

1. INTRODUCTION

It has long been confirmed that many disc galaxies present more complexity than what may first meet the eye. In our own Milky Way, the pioneering discovery of 21-cm HI emission (Ewen & Purcell; Muller & Oort 1951) gave way to the possibility of structural mapping far beyond the visible stellar disc, and since then various complexities of the disc have been recognised. Of these, a warp, a vertical undulation causing systematic vertical displacement of the disc from the mid-plane, has been one of the most interesting to study. First established through neutral hydrogen (HI) mapping (Kerr 1957), its geometry and motion have since been mapped extensively through both gaseous and stellar tracers (e.g., Poggio et al. 2018; Jónsson & McMillan 2024; Khanna et al. 2025).

Observations of external galaxies have also revealed that the phenomenon of warping is almost ubiquitous, being observed in 40-50% of edge-on spiral galaxies (Richter & Sancisi 1994; Reshetnikov & Combes 1998), several of which exhibit similar features seen in our own galaxy (Bottema 1995; Vollmer et al. 2016). These findings imply that the Milky Way is not atypical in its warped morphology; rather, perhaps that warping is a natural outcome of disc galaxy evolution. While the origin of warping within our Galactic disc is still an open ques-

tion, current leading theories include a misaligned or rotating dark matter halo (Han et al. 2023), accretion of gas with angular momentum not aligned with the Galactic disc (Jiang & Binney 1999), or tidal interactions between the Milky Way and companion galaxies (Khanna et al. 2019; Laporte et al. 2019). Progress has been made in ruling out non-gravitational origins of the warp by Poggio et al. 2018. By comparing the kinematics of young upper main sequence (UMS) stars with those of evolved giants, warp signatures were identified in both samples. This does not align with models in which the warp is excited by non-gravitational mechanisms, such as magnetic fields or hydrodynamic pressure from infalling gas, which primarily affect the gaseous disc and younger stellar populations (Guijarro et al. 2010; Sellwood 2013). Younger stars generally trace the kinematics of the gas disc from which they formed (McKee & Ostriker 2007), not having had enough time to be dynamically heated by past perturbations of the disc. Therefore, the measured global warp signature among the range of samples points toward the warp being a gravitationally induced phenomenon.

Furthermore, the spatial correlation between the warp and the radial flaring of the disc, where the scale height begins to increase with Galactocentric radius, implies that the warp is not a static feature

but one that is dynamically evolving with time (Upal et al. 2023). This state of non-equilibrium is a constraint of theoretical models, and the gravitational mechanisms previously discussed successfully reproduce such warps, reinforcing their feasibility as the mechanisms that may have originally excited the disc.

In order to further elucidate the origin of the Milky Way’s warp, it is important to understand its history and formation. Being the galaxy in which we reside, the Milky Way provides the closest and most detailed example of a warp that we can study. Recent radio surveys have allowed for detailed mapping of the outer gaseous layers of the Galactic disc (Han et al. 2021; Pingel et al. 2023), while photometric surveys such as 2MASS and AllWISE (Skrutskie et al. 2006; Cutri et al. 2013) have allowed for precise measurements of millions of stars. However, while gas warping is well documented, this paper will focus on the warp found in the stellar component of the disc.

Stellar tracers have been fundamental in tracing the structure and kinematics of the warp, with individual tracers providing unique advantages. Classical Cepheids, being young stars (ages < 500 Myr; Catelan & Smith 2015), inherit the motions of the neutral hydrogen gas where they have formed, largely unaffected by dynamical perturbations or heating that obscure the initial orbital characteristics of older stellar populations. Because of this, they trace the young stellar disc and spiral arm architecture (Skowron et al. 2019; Cabrera-Gadea et al. 2024). Conversely, red clump (RC) stars represent the intermediate-to-old stellar populations (ages $\sim 1\text{--}10$ Gyr; Bovy et al. 2014), and due to their age, exhibit more chaotic orbits caused by past disturbances (Poggio et al. 2018). RC stars are abundant across the Galaxy and therefore provide the high density of data points required for the mapping of the large-scale kinematics of the Milky Way.

By synthesising the data from these recent missions and use of stellar tracers, we have advanced our understanding of the geometry of the Milky Way’s warp. The warp exhibits a complex, asymmetric morphology with disc vertical displacements that can be described to increase as a power-law function of the Galactocentric radius (Zhou et al. 2025). It has also been revealed that the warping is likely not a simple bend in the disc but is characterised by a ‘twisted’ line of nodes (hereafter LoN) that rotates with increasing Galactocentric radius, presenting a leading spiral pattern that may suggest

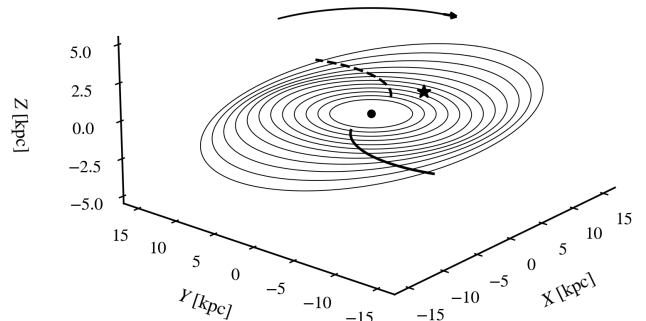


FIG. 1.— Three-dimensional schematic of the Milky Way’s warped stellar disc. The schematic is constructed from tilted concentric rings with the warp amplitude increasing beyond a Galactocentric radius of 10 kpc. A twisted LoN is illustrated by the solid and dotted black lines representing the ascending and descending nodes, respectively. The Galactic centre (dot) and the Sun (star) are shown for reference, with the arrow showing the direction of stellar rotation.

a system evolving over time (Jónsson & McMillan 2024). The LoN represents the line in which the stellar disc of the Galaxy intersects the flat mid-plane, marking the transition where the disc crosses upward or downward at the ascending and descending nodes, respectively. Figure 1 illustrates a twisted LoN on a 3D Milky Way model.

Despite the utility of these stellar tracers, tracing the geometry and kinematics of the Milky Way is still a challenge due in part to the layer of dust seen in the Galactic mid-plane. Arising from the cool outflows of dying stars that mix with the dense layer of neutral hydrogen concentrated at the mid-plane, this extinction layer obscures the view to many stars, particularly toward the inner Galaxy and the far side of the disc. To minimise the loss of stellar data within catalogues, progress has been made in creating extensive 3D extinction maps to understand the number and positions of stars obscured by dust (Khanna et al. 2025). By analysing wavelength-dependent absorption of starlight, these maps allow for detailed and comprehensive mapping of the warp in the areas of the Galaxy where light is blocked by dust. It must be noted that while this mid-plane dust obscures the view to many stars, it does not bias the measured kinematics of the stars that can still be observed.

Figure 2 visualises regions where certain lines of sight are affected by extinction. Across the three panels, which represent increasing Galactocentric radii, a clear signature of warping is seen: the stellar distribution at lower azimuths rises above the mid-plane with higher azimuths sinking. Consequently, the red line representing the median vertical position displays fluctuations that likely arise

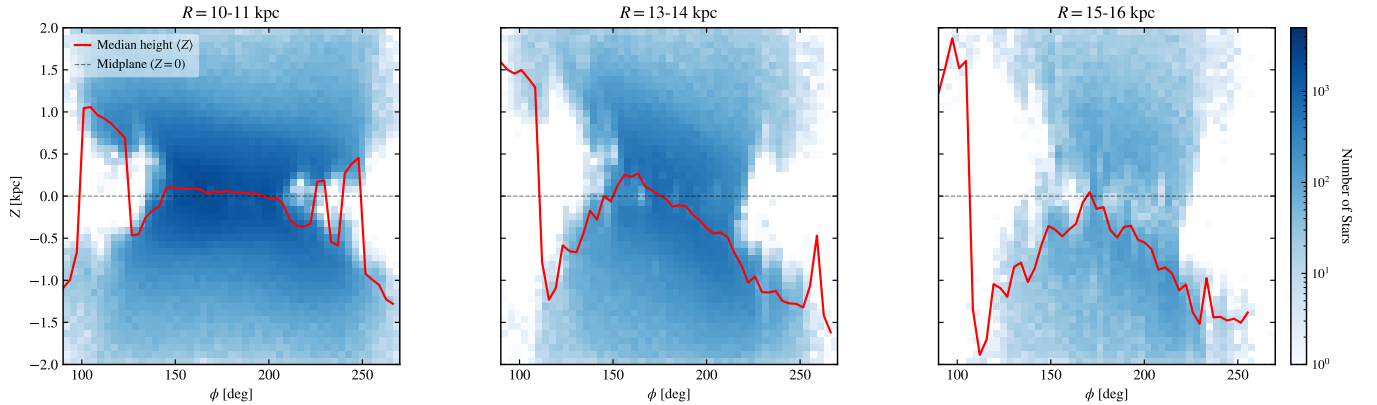


FIG. 2.— Vertical distribution of RC stars within the dataset with respect to Galactocentric azimuth. Panels from left to right show radial slices at $R = 10 - 11$ kpc, $13 - 14$ kpc, and $15 - 16$ kpc. The red line indicates the median stellar vertical position (Z), with the colour intensity representing the stellar number density. Distinct fluctuations of the median height of the stellar sample are explained by artificial skewing arising from high dust extinction obscuring the view to stars along specific line of sights. The grid resolution is $\sim 4^\circ$ in azimuth and 80 pc in vertical height.

due to observational bias, areas of stars obscured by mid-plane dust, creating extinction-induced artifacts that do not represent the actual star distribution throughout the Galaxy.

For a simple, static warp, vertical velocities would be predicted to coincide with stars with maximal vertical displacement. However, recent mapping has shown that the vertical velocities measured do not line up with the maximum height of the warp (Poggio et al. 2018). The velocity pattern is shifted, behaving like a wave that is slightly ahead of the crest of displacement. This phase shift is the signature of a bending wave that is moving through the disc, and from this, a differential precession rate in the prograde direction of $10\text{--}14 \text{ km s}^{-1} \text{ kpc}^{-1}$ was calculated (Poggio et al. 2020; Jónsson & McMillan 2024; Cheng et al. 2020). The vertical velocity field describes a systematic wobble of stars perpendicular to the disc; there is a clear gradient where v_z increases by roughly $5\text{--}6 \text{ km s}^{-1}$ increasing from $8\text{--}14$ kpc in Galactocentric radius (Poggio et al. 2020; Gaia Collaboration et al. 2023).

To address the uncertainties still remaining within the warp, this study will use a sample of RC stars from *Gaia* DR3 (Gaia Collaboration et al. 2023) and AllWISE (Cutri et al. 2013). The data are fitted to two different rigid precessing disc models: one with a fixed LoN and precession rate across the disc, and one that allows them to vary linearly with Galactocentric radius. Markov Chain Monte Carlo (MCMC) analysis is performed on the models to find best fitting parameters for warp height at varying radii, precession rate of the warp as well as the warp’s LoN. These results are compared to known literature to better understand the evolution of this

dynamic structure and better constrain the mechanism that excited the warp, and allow for a more definitive distinction between long-lived structural features and transient, impulse-like perturbations.

2. DATA

The dataset used for this study is a sample of RC stars from the official cross-match between *Gaia* DR3 and AllWISE; details of the cross-match algorithm can be found in Marrese et al. 2019. Only stars with finite measurements for parallax (ϖ) and photometry in both *Gaia* (G) and AllWISE (W_1 and W_2) bands were kept. The combination of these datasets is especially effective: AllWISE’s mid-infrared photometry penetrates the obscuring mid-plane dust, while *Gaia* provides precise astrometry and distance estimates. RC stars are intermediate-to-old low-mass stars that have exhausted the hydrogen in their cores and have undergone a helium flash to ignite helium burning. Because helium ignites at near-identical core masses, they exhibit similar absolute magnitudes, making them excellent standard candles for distance estimation and exceptional tracers for understanding the structure of the Galaxy and its warp (Girardi 2016).

The selection of RC stars from the initial cross-match sample of $N = 303\,183\,850$ began with the calculation of their absolute magnitudes. To avoid large errors when inverting uncertain parallaxes, Bayesian distance estimates were employed from Bailer-Jones et al. 2021. These distance priors were combined with 3D extinction maps (Lallement et al. 2019; Green et al. 2019) to estimate the foreground extinction along the line of sight, yielding a posterior probability distribution of the inferred absolute magnitude in both the G and W_1 bands.

To select the final RC candidates, a probabilistic method was applied by treating the true absolute magnitude of RC stars as a 2D Gaussian distribution based on literature (Hawkins et al. 2017). A Mahalanobis distance (Mahalanobis 1936) was calculated between the star’s inferred absolute magnitude and the expected RC absolute magnitude, with stars classified as RC candidates if their log-likelihood exceeded the threshold that corresponds to a ‘perfect’ RC star placed at a Mahalanobis distance of 3. A more detailed explanation of the selection process is described in Khanna et al. 2025, and the process resulted in a sample of 10 269 109 RC stars.

To ensure the high quality of the RC sample, several masks and quality cuts were applied to the data. First, all sources within a Galactocentric radius of $R < 3$ kpc were removed to avoid the complex kinematics and contamination of the Galactic bulge and central bar. Second, sources beyond a vertical distance of 2 kpc from the Galactic mid-plane ($|Z_{GC}| > 2$ kpc) were removed to exclude contamination from high-latitude sources in the Large and Small Magellanic Clouds. Third, the worst 0.5% of the data was excluded by requiring the photometric uncertainty in the AllWISE W_1 band to be $\sigma_{W_1} < 0.1$. Finally, a faint-end magnitude limit of $G = 17.5$ was imposed. These cuts generated a high-quality sample of 8 729 562 RC stars spatially centred on the Sun’s position within the Galaxy.

2.1. Line-of-Sight Velocity Estimates

To analyse the Galactocentric vertical velocity for the full sample, full 3D kinematic coordinates must be known, and this is the case for 52% of the dataset. However, due to the higher signal-to-noise ratio required to measure the Doppler effect, 48% of the dimmer stars within the sample lacked these velocities. To avoid further reducing the sample size for modelling, missing line-of-sight velocities were derived under the assumption that the Galactocentric radial velocity is zero at any point in the disc ($v_{R,gal} = 0$). While these radial velocities can be significant in localised regions, it has been shown that they are small in comparison to the rotational velocity, and therefore the estimation is successful on larger scales (McMillan et al. 2022). Typical Galactocentric radial velocities of $v_{R,gal} \approx 10$ km s^{−1} are found in the extended solar neighbourhood (Gaia Collaboration et al. 2018), which contrast with much larger Galactocentric rotational velocities of order $v_{\phi,gal} \approx 200$ km s^{−1} (Reid et al.

2019).

The Sun’s Galactocentric cylindrical radius is adopted as $R_{\odot} = 8.178$ kpc (GRAVITY Collaboration et al. 2019) and its height above the mid-plane as $z = 20.0$ pc (Humphreys & Larsen 1995), along with velocities $\vec{v}_{\odot} = (v_{R,\odot}, v_{\phi,\odot}, v_{z,\odot}) = (11.1, 241.2, 7.25)$ km s^{−1} (Schönrich et al. 2010; Eilers et al. 2019). In Cartesian coordinates, the true Galactocentric velocity of a star, $\vec{v}_{GC}$, is the vector sum of its motion derived from its proper motion and the motion derived from its line-of-sight velocity. Letting $\vec{v}_{GC,0}$ be the Galactocentric velocity vector under the assumption that $v_{los} = 0$, the true Galactocentric velocity can be expressed as a linear combination of this baseline velocity and the contribution from the true line-of-sight velocity, given by:

$$\vec{v}_{GC} = \vec{v}_{GC,0} + v_{los}\hat{n} \quad (1)$$

where $\hat{n} = (n_x, n_y, n_z)$ is the unit vector pointing from the Sun to the star. In the Galactocentric cylindrical frame, the planar radius vector is $\vec{R} = (x, y)$, and the associated unit vector in the radial direction is $\hat{e}_R = (x/R, y/R, 0)$, where $R = \sqrt{x^2 + y^2}$. Therefore, the Galactocentric radial velocity, v_R , is the projection of the total Galactocentric velocity onto $\hat{e}_R$:

$$v_R = \vec{v}_{GC} \cdot \hat{e}_R = v_{R,0} + v_{los}B \quad (2)$$

where $v_{R,0} = \vec{v}_{GC,0} \cdot \hat{e}_R$ is the Galactocentric radial velocity contribution from proper motions and solar motion. The parameter B is a geometric projection factor defined as the dot product of the line-of-sight unit vector and the Galactocentric radial unit vector. By imposing the assumption that the star follows a circular orbit ($v_R = 0$), the equation can be rearranged to solve for the unknown line-of-sight velocity,

$$v_{los} = -\frac{v_{R,0}}{B}. \quad (3)$$

The stability of the v_{los} estimation is dependent on the projection factor, B , leading to specific regions in which this assumption works exceptionally well. For example, when looking directly toward or away from the Galactic centre ($l = 0^\circ, 180^\circ$), the Galactocentric radius aligns with the line of sight, allowing the projection factor to reach its maximum absolute value ($|B| \approx 1$). In contrast, in regions looking toward or away from the direction of Galactic

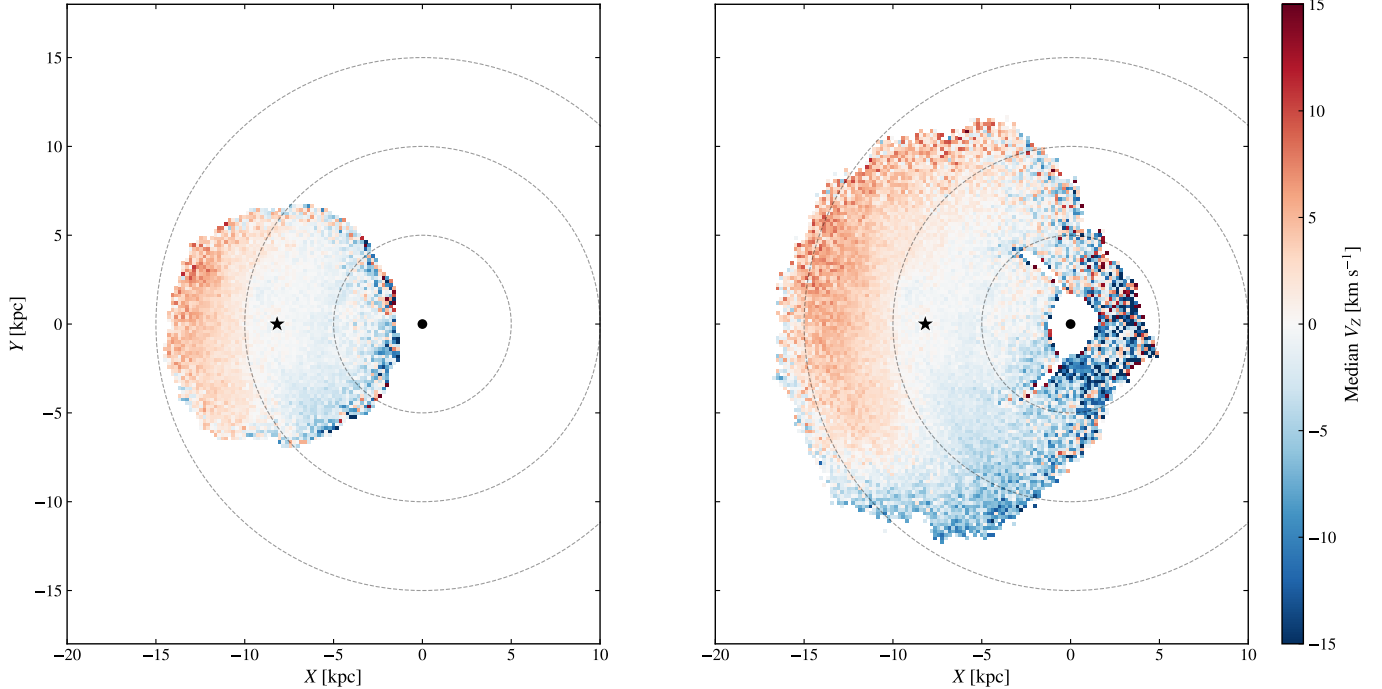


FIG. 3.— Top-down median vertical velocity distribution (v_z) for RC sample in Cartesian coordinates (the Sun and Galactic centre are indicated; Galactocentric circles differ by 5 kpc in radius). The left panel displays the spatial coverage of stars with complete 3D velocity data. The right panel shows the expanded spatial footprint when including estimated line-of-sight velocities from the assumption of zero Galactocentric radial velocity ($v_{R,gal} \approx 0$). Bins contain a minimum of 50 stars and any bins with fewer than 50 stars were not plotted.

rotation ($l = 90^\circ, 270^\circ$), the line of sight is perpendicular to the Galactocentric radial vector, and the projection factor approaches zero. To mitigate the division by values close to zero, which results in line-of-sight velocities approaching infinity, a geometric filter was applied, and estimations were only accepted for stars where the projection factor satisfied $|B| > 0.1$. Furthermore, it must be noted that the calculation experiences physical breakdown in regions where the assumption of zero Galactocentric radial velocity is violated. Toward the Galactic centre, the presence of the Galactic bar heavily perturbs nearby orbits, inducing high eccentricities. Similarly, at high Galactic latitudes, this planar model projects 3D sight lines onto a 2D kinematic constraint. For stars at large distances away from the mid-plane ($|b| \gg 0^\circ$), the sight line introduces a steep angle, causing $\hat{n}$ to become increasingly orthogonal to the planar vector $\hat{e}_R$ and resulting in an overestimation of the line-of-sight velocity. A detailed discussion of the uncertainties introduced by using this approximation is presented in Appendix A.

Even with these limitations, this assumption of zero Galactocentric radial velocity provides accurate line-of-sight velocities for 3 860 894 sources, increasing the 6D phase-space coordinate coverage of the sample to 96%. The resulting increase in spatial

coverage is illustrated in Figure 3.

3. DISC MODELS

3.1. First-Order Fourier Mode

The warp models described here have been adopted from Jónsson & McMillan 2024, with some additions made to the twisted model. They assume a rigidly precessing disc where the contribution of radial motion to vertical velocity is negligible. The vertical position of stars in Galactocentric cylindrical coordinates depends on parameters R and ϕ (Cheng et al. 2020; Jónsson & McMillan 2024) such that:

$$z_0(R, \phi) = h(R) \sin(\phi - \phi'_w(R) - \omega_p(R)t), \quad (4)$$

where t is time, $h(R)$ is the largest displacement of the warp at a given Galactocentric radius (R), ϕ is the Galactocentric azimuth, $\phi'_w(R)$ is the LoN, and $\omega_p(R)$ is the precession rate of the warp.

To model the variation in the warp height, a piecewise-linear model was employed rather than a single continuous equation to describe the entire disc; the radii are split into 1 kpc intervals from $R = 7$ to 16 kpc (denoted as parameters $h_7, h_8, \dots, h_{16}$). This allows for a higher flexibility of the model and avoids forcing a specific assumed shape.

Two specific configurations of this warp shape are

tested: the first is the fixed model, where both the LoN (ϕ'_w) and the precession rate (ω_p) stay constant across the entire disc. The second configuration allowed both of these parameters to vary linearly with Galactocentric radius and is denoted as the twisted model, accommodating a ‘twisted’ warp where the LoN and precession rate are defined as:

$$\phi'_w(R) = \phi'_{w,0} + \phi_{w,1}(R - R_s) \quad (5)$$

$$\omega_p(R) = \omega_{p,0} + \omega_{p,1}(R - R_s), \quad (6)$$

where R_s is the reference radius placed at 10 kpc as a representative midpoint for the outer-disc data. Both $\phi'_{w,0}$ and $\omega_{p,0}$ are therefore the base LoN and precession rate at this reference radius, with $\phi_{w,1}$ and $\omega_{p,1}$ representing their gradient parameters, describing how they vary with respect to Galactocentric radius. To improve upon the models seen in [Jónsson & McMillan 2024](#) and increase the physical accuracy of the twisted model, constraints were placed on where and how much certain parameters could vary. Firstly, because no significant warping is observed in the inner disc ($R < 10$ kpc), the LoN and precession rate were constrained to vary only at radii beyond 10 kpc. Secondly, the precession gradient was constrained to be within $\pm 2 \text{ km s}^{-1} \text{ kpc}^{-2}$. Finally, the warp height was forced to be increasing monotonically with Galactocentric radius. These constraints on the twisted model prevent the Markov Chain Monte Carlo algorithm from exploring mathematically correct yet unphysical solutions.

3.2. Expected Vertical Velocities

To then calculate expected vertical velocities for these models, Jeans’ first equation was used ([Jeans 1915](#)) as a fluid continuity equation to describe the collisionless, high density particle system that is assumed in these models. The equation in cylindrical polar coordinates can be written as:

$$\frac{\partial n}{\partial t} + \frac{\partial(n\bar{v}_R)}{\partial R} + \frac{1}{R} \frac{\partial(n\bar{v}_\phi)}{\partial \phi} + \frac{\partial(n\bar{v}_z)}{\partial z} = 0, \quad (7)$$

where n is the stellar number density, assumed to follow a double exponential profile defined as

$$n \propto \exp\left(-\frac{|z'|}{z_h}\right) \exp\left(-\frac{R}{R_h}\right), \quad (8)$$

where $z' = z - z_0(R, \phi)$ and the scale height (z_h) and scale length (R_h) are not included in the final velocity equations.

Assumptions are made to the kinematics such as that azimuthal velocity does not vary with azimuth ($\partial \bar{v}_\phi / \partial \phi = 0$), and that vertical velocity does not vary with height above or below the mid-plane ($\partial \bar{v}_z / \partial z = 0$). Applying the product rule, the kinematic assumptions, and the chain rule to Equation 7 isolates the mean vertical velocity:

$$\bar{v}_{z,\text{mod}} = \left(\frac{\bar{v}_\phi}{R} - \omega_p(R) \right) h(R) \times \frac{\partial}{\partial \phi} \sin\left(\phi - \phi'_w(R) - \omega_p(R)t\right) \quad (9)$$

Equation 9 describes the mean expected vertical velocities for the fixed and twisted model used in this study to fit the observed and estimated data. It must also be noted that the LoN found by the fitting algorithm generally favoured the warp on the opposing side of the disc; therefore, 180° is added to the calculated LoN for all models to represent the symmetric side of the warp that is closer to the Sun.

3.3. Fitting the Data

In order to fit the data to the described models, a posterior probability density function (PDF) must be found. Direct analytical calculations of these functions with many parameters and points become increasingly prohibitive due to the exponential increase in computational cost as dimensionality increases. Therefore, to approximate the posterior PDF the use of the Markov chain Monte Carlo (MCMC) algorithm is employed, specifically the `emcee` Python package ([Foreman-Mackey et al. 2013](#)).

`emcee` is an affine-invariant ensemble sampler for MCMC. The adaptation from base MCMC analysis provides significant advantages over standard algorithms, such as the Metropolis-Hastings (M-H), when working with many-parameter models, and consequently has been utilised for many research applications in cosmology and astrophysics. The goal of these algorithms is to sample from the posterior PDF, defined using Bayes’ theorem as:

$$p(\Theta, \alpha | D) = \frac{1}{Z} p(\Theta, \alpha) p(D | \Theta, \alpha), \quad (10)$$

where the prior function $p(\Theta, \alpha)$ and the likelihood function $p(D | \Theta, \alpha)$ are not computationally expensive to evaluate for any given useful parameter Θ_i or

nuisance parameter α_i , where a nuisance parameter is one that is required for modelling but is not of primary scientific interest. It is the normalisation constant, Z , that becomes impossible to compute for large models and is therefore not required during MCMC analysis due to the use of ratios between different steps as seen in Equations 13 and 14. To reduce computational overhead, nuisance parameters are marginalised with the integral

$$p(\Theta|D) = \int p(\Theta, \alpha|D) d\alpha, \quad (11)$$

effectively treating the nuisance parameters as an uncertainty that can be included in the final result. It must be noted that the specific kinematic models used for this study do not contain any nuisance parameters, and therefore the posterior probability ($p(\Theta|D)$) depends entirely on the variables of scientific interest. At the end of the MCMC process, the true expectation value of a function ($\langle f(\Theta) \rangle$) is calculated to be the approximate average of all M independent samples:

$$\langle f(\Theta) \rangle \approx \frac{1}{M} \sum_{i=1}^M f(\Theta_i). \quad (12)$$

It is in the movement of the algorithm through parameter space that the advantages of **emcee** can be seen. In the traditional M-H method the chance of accepting a move from an initial position ($X(t)$) to a new position (Y) is calculated with

$$\min \left(1, \frac{p(Y|D)}{p(X(t)|D)} \frac{Q(X(t); Y)}{Q(Y; X(t))} \right). \quad (13)$$

Here, the probability is calculated with the comparison of the posterior probabilities between the two positions weighted by the proposal distribution Q . In contrast, **emcee** chooses a new position of a walker (X_k) not weighted by a chosen distribution but along the line connecting it to a partner walker (X_j), with the distance along this line determined by a random variable Z . This requires a new acceptance rule that is similar to Equation 13 but includes a factor of Z^{N-1} , where N is the number of dimensions in parameter space.

$$q = \min \left(1, Z^{N-1} \frac{p(Y)}{p(X(t))} \right). \quad (14)$$

It is because of this change in movement through parameter space that makes **emcee** incredibly efficient

at finding posterior PDFs for many parameter models. However, because walkers are initialised without existing knowledge of the PDF, they undergo a ‘burn-in’ phase, during which they reach a stationary distribution from arbitrary initial positions. To discard this phase, during modelling the first 20% of steps are discarded to account for ‘burn-in’ time and only include samples in which the walkers are exploring regions of high likelihood.

To evaluate the posterior probability at each step, a log-likelihood function must be defined. Here, flat priors were assumed for all parameters within physically allowed regions. The total uncertainty for each spatial bin, σ_{tot} , was calculated by adding the statistical bootstrapping uncertainty (σ_{stat}), a constant systematic uncertainty (σ_{sys}) of 0.7 km s^{-1} , and the propagated model uncertainty (σ_{mod}) in quadrature:

$$\sigma_{tot} = \sqrt{\sigma_{stat}^2 + \sigma_{mod}^2 + \sigma_{sys}^2}. \quad (15)$$

The log-probability sampled by the walkers is therefore equivalent to the log-likelihood, defined as:

$$\ln p \propto -\frac{1}{2} \sum \frac{(\bar{v}_{z,mod} - \bar{v}_{z,data})^2}{\sigma_{tot}^2}. \quad (16)$$

where the sum is calculated over all spatial bins included in the sample.

The total number of parameters modelled using **emcee** includes 12 for the fixed model: warp height in 1 kpc sections from radii 7 to 16 kpc ($h_7, h_8, \dots, h_{16}$), LoN (ϕ'_w), and precession rate (ω_p). The twisted model adds two more parameters to this: specifically, the LoN and precession rate gradients ($\phi_{w,1}$ and $\omega_{p,1}$ respectively). These 12 and 14 parameter models correspond to the dimensionality $N = 12$ and $N = 14$ in Equation 14.

To find these best fitting parameters, **emcee** was used with 50 walkers exploring this parameter space over 20 000 steps to ensure that there was sufficient time to arrive at confident results.

4. RESULTS

4.1. Model-Data Comparison

Once MCMC analysis has been performed on the models and compared to the dataset, the fit to the data must be analysed to understand if the vertical velocity trends have been reproduced successfully. This fit can be seen in Figure 4 where both models successfully reproduce the vertical velocity trends associated with a warping disc. Regions where both models closely match the data fall between 140°

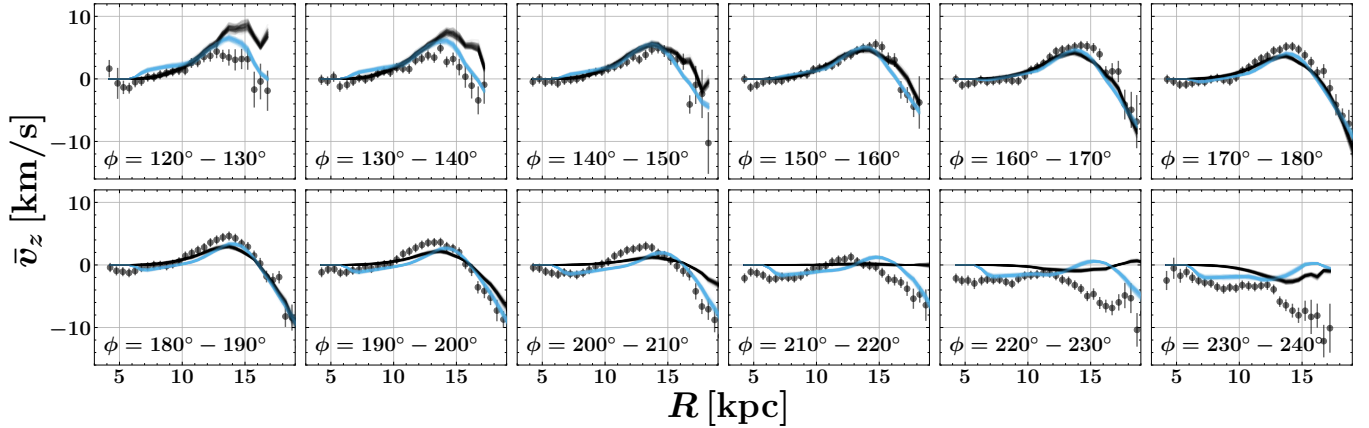


FIG. 4.— Fit of the expected vertical velocities ($\bar{v}_z$) of the twisted model (blue lines) and fixed model (black lines) to the dataset (black dots) as a function of Galactocentric radius (R) analysed in bins of 10 degrees Galactocentric azimuth (ϕ). Observed data contains errors derived with bootstrapping.

and 200° , with angles outside of this exhibiting discrepancies between expected vertical velocities and observed. Discrepancies can be expected due to the simplicity of the model and its failure to account for vertical velocities that may have arisen from other mechanisms. Another feature that both models portray is an overestimation of velocities at low azimuths and underestimation at high azimuths. This is an expected feature and one that signifies a warp that cannot be entirely modelled using first-order Fourier models. One limitation of the fixed and twisted models, being of first-order, is that they force a strict global antisymmetry. Studies such as [Levine et al. 2006](#) and [Cabrera-Gadea et al. 2024](#) have found that the Milky Way’s warp is lopsided and exhibits different warp amplitudes on the opposite sides, and therefore forcing an antisymmetric shape can lead to the model attempting to average out the two warp amplitudes, causing discrepancies at extreme azimuths.

Outside of these limitations, in these regions of extreme azimuths the twisted model generally traces closer to the data. Where the fixed model flattens out near $\phi \approx 210^\circ - 230^\circ$, the twisted model, by allowing the LoN and precession to vary, introduces a phase shift in the outer disc. This twist allows more accurate downward velocity reproduction and may help provide evidence that the warp twists with Galactocentric radius.

Another feature that can be seen in Figure 4 is the line of maximum vertical velocity (LMV_z). Equations 4 and 9 dictate that the LMV_z will occur exactly at the LoN (approximately $\phi \approx 130^\circ - 150^\circ$ in Figure 4). This result does not agree with the offset found between the LoN and LMV_z found in [Cabrera-Gadea et al. 2024](#) and [Poggio et al. 2025](#),

such a phase shift between these two lines suggests a warp that is not rigidly precessing but propagating through the disc as a transient bending wave.

4.2. Warp Shape

The resulting warp shapes from these models varied in amplitude and shape. The top panel in Figure 5 displays the resulting warp shapes for the first-order Fourier models. It demonstrates a successful reproduction of a warping outer disc with the two independent models varying in the amplitude of the warp in the outer regions of the disc. For comparison, also plotted is the model found by [Cheng et al. 2020](#) using kinematic data from *Gaia* DR2, StarHorse and the APOGEE survey. The bottom panel displays comparative warp shapes from recent literature that use photometric star data ([López-Corredoira et al. 2002](#); [Amôres et al. 2017](#); [Cheng et al. 2020](#); [Chrobáková et al. 2020](#); [Romero-Gómez et al. 2019](#); [Uppal et al. 2024](#)).

The fixed model displays steep bending of the outer disc reaching a maximum amplitude of 4.1 ± 0.2 kpc at a Galactocentric radius of 16 kpc. The warp shape for this model is most similar to the one defined in [Amôres et al. 2017](#), exhibiting a stronger warp than the other chosen comparison models.

Comparing this to the twisted model, here a more moderate warp shape is estimated reaching only 1.13 ± 0.1 kpc at 16 kpc away from the Galactic centre. Additionally, beyond $R = 14$ kpc the twisted model plateaus, a feature that is also found in [Cabrera-Gadea et al. 2024](#), however in that study plateauing is only measured between $10 < R/\text{kpc} < 11$ and therefore is not representative of the shape seen in Figure 5. An interesting feature of Figure 5 is the relationship between the twisted model found

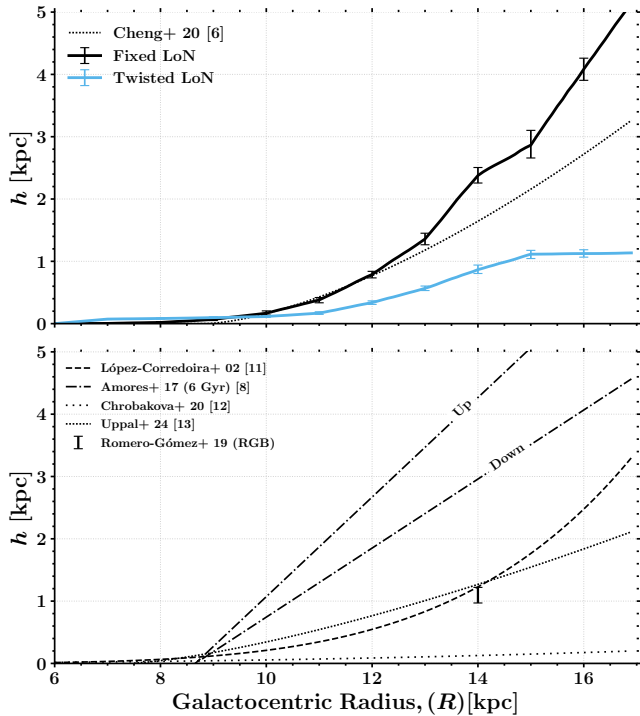


FIG. 5.— Top panel: Maximum warp amplitude as a function of Galactocentric radius for the fixed and twisted models. Solid lines represent the median values derived from the models’ MCMC posteriors, with the error bars showing the 1σ confidence intervals. The twisted LoN (blue line) displays a moderate warp reaching a maximum amplitude of 1.13 ± 0.07 kpc at 16 kpc Galactocentric radius, while the fixed LoN model (black line) reflects much steeper disc bending with a deviation of 4.11 ± 0.20 kpc from the mid-plane at 16 kpc away from the Galactic centre. Bottom Panel: Comparative warp shapes reconstructed from various recent literature findings. Figure adapted from diagram in Jónsson & McMillan 2024.

here and the model found in López-Corredoira et al. 2002 also using RC stars. Similar amplitudes are observed up to $R = 14$ kpc, however, beyond this point they diverge: the shape found in this study plateaus while the warp seen in López-Corredoira et al. 2002 continues to grow exponentially. Similarly, Uppal et al. 2024 also uses RC stars and finds a continuous growth of the disc in the outer regions; however, here the warp amplitude is greater, reaching ~ 1.4 kpc at $R = 14$ kpc.

Beyond the comparison models shown in Figure 5, the twisted model exhibits a similar warp amplitude at $R = 14$ kpc to that found by Marshall et al. 2006 using dust as a tracer and a largely similar warp amplitude found by Lemasle et al. 2022 using younger Cepheids. Conversely, the amplitude found by Romero-Gómez et al. 2019 using RGB stars is consistently greater than that found by the twisted model here, finding a height at $R = 14$ kpc of 0.97 kpc and -1.22 kpc for the Northern and Southern sides, respectively. Interestingly, Cabrera-Gadea et al. 2025 using the older stellar population

of RR Lyrae stars finds a warp amplitude that is systematically lower than the Cepheid warp found in Cabrera-Gadea et al. 2024.

4.3. Precession Rate

The rates of precession of the warp around the galaxy found by the fixed and twisted models are displayed in Figure 6. The fixed model (black line) reaches a confident result of -12.4 ± 0.1 $\text{km s}^{-1} \text{kpc}^{-1}$ in the direction of stellar rotation with only small variation in the walkers during the MCMC process; this suggests that the precession rate found in this model along with the other parameters fit accurately to the data and the resulting parameters are robust.

This fixed prograde precession rate across the disc largely agrees with studies using *Gaia* DR2 (Cheng et al. 2020; Poggio et al. 2020), with the exception of Chrobáková & López-Corredoira 2021, who find a much slower precession rate (4^{+6}_{-4} $\text{km s}^{-1} \text{kpc}^{-1}$) and do not rule out a non-precessing warp. *Gaia* DR3 studies have also generally agreed on a prograde precession rate, with Jónsson & McMillan 2024 and Cabrera-Gadea et al. 2025 finding similar rates of ~ 13 $\text{km s}^{-1} \text{kpc}^{-1}$. However, studies using classical Cepheids result in a more moderate rate of around $\sim 4 - 7$ $\text{km s}^{-1} \text{kpc}^{-1}$ (He 2023; Zhou et al. 2025).

The twisted model, however (blue line), does not reach the conclusion of a fixed precession rate across the disc but finds one that varies with Galactocentric radius. In the inner disc ($R < 10$ kpc), a prograde precession rate of -0.9 $\text{km s}^{-1} \text{kpc}^{-1}$ is found due to the constraint of the precession rate gradient, which is enforced because the small warp amplitude in this region makes accurate changes in the precession rate difficult to calculate (Dehnen et al. 2023). Beyond this region, where the precession rate is allowed to vary, it increases in the prograde direction to a rate of -1.9 ± 0.1 $\text{km s}^{-1} \text{kpc}^{-2}$ reaching a rate of -12.3 $\text{km s}^{-1} \text{kpc}^{-1}$ at $R = 16$ kpc. A precession rate that varies across the Galactic disc is also well represented in recent studies. Cabrera-Gadea et al. 2024 find a slight increase of warp precession rate in the prograde direction in the outer disc, while Huang et al. 2024 also finds a varying precession rate but one that begins in the retrograde direction, both of these studies using Cepheids. Conversely, Dehnen et al. 2023, also using Cepheids, finds a decrease in prograde precession with respect to Galactocentric radius ranging from ~ 12 $\text{km s}^{-1} \text{kpc}^{-1}$ at $R = 12$ kpc to ~ 6 $\text{km s}^{-1} \text{kpc}^{-1}$ at $R \geq 14$ kpc.

From the results of both of these models, the warp

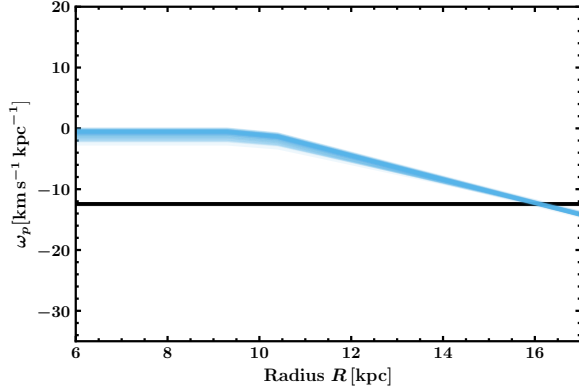


FIG. 6.— Comparison between the precession rates of the fixed model (black line) and the twisted model (blue line) against Galactocentric radius. Both lines are constructed from 100 random realisations of the MCMC posterior distribution.

in the Milky Way is undergoing precession and propagating rapidly through the disc. Moreover, the varying precession found in the twisted model suggests that the warp is experiencing differential precession. Outer regions of the disc precessing at a faster rate compared to inner regions suggests that the warp itself will twist into a spiral and therefore ‘wind up’.

4.4. Line of Nodes

The fixed model results in a LoN of $125.4^\circ \pm 1.1^\circ$, a 55° prograde offset from the Sun’s Galactic azimuth. The twisted model, however, exhibits a trailing spiral beginning at 98.5° at $R = 10$ kpc and reaching 149.6° Galactocentric azimuth at $R = 16$ kpc.

A fixed angle is favoured by a few studies using Cepheids (Chrobáková et al. 2020; Huang et al. 2024) placing the LoN near 175° , with similar results from Momany et al. 2006 and Amôres et al. 2017 using older stellar samples. Additionally, Huang et al. 2024 measure a time-dependent but fixed LoN that precesses at a rate of $-2.1 \pm 0.5 \text{ km s}^{-1} \text{ kpc}^{-1}$ in the retrograde direction using Cepheids. In contrast, recent studies have also found that the LoN is not fixed but varies with Galactocentric radius, similar to what is shown by the twisted model. Cabrera-Gadea et al. 2024 find no significant change of the LoN within the inner galaxy ($R < 11$ kpc), however beyond this the line twists in the direction of galactic rotation at a rate of $-12.7 \pm 0.3^\circ \text{ kpc}^{-1}$; the same leading spiral pattern was also found by other studies using Cepheids (Chen et al. 2019; Dehnen et al. 2023; Zhou et al. 2025), although each varying at different rates. Results finding a leading spiral would agree with Briggs’ third rule for Galactic warps (Briggs

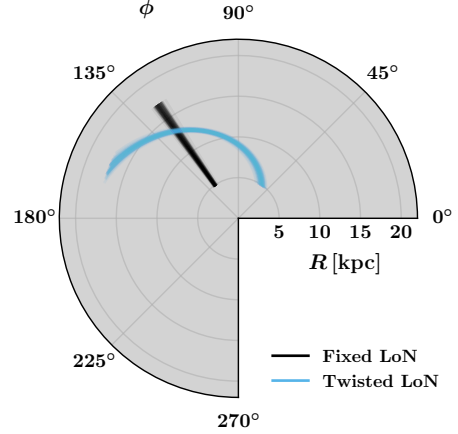


FIG. 7.— Top-down representation of the Galactic warp’s Line of Nodes (LoN) as a function of azimuthal angle (ϕ) against Galactocentric radius (R) constructed from 100 random realisations of the MCMC posterior. The fixed model (black) exhibits a constant line of nodes across the entire disc, while the twisted model (blue) reveals a LoN that resembles a trailing spiral.

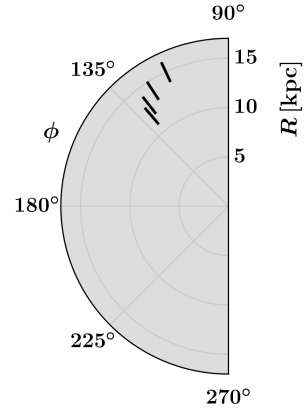


FIG. 8.— Top-down representation of the warp’s LoN from the fixed model after splitting data into 2 kpc annuli from 11-13 kpc to 14-16 kpc. The pattern here reveals a leading spiral pattern that contrasts with the global fit of one that is trailing.

1990) but go against what is calculated from the twisted model shown in Figure 7.

Importantly, when the dataset is split into 2 kpc annuli and tested with the fixed model, a spiral pattern is found that opposes the direction of the one derived from the twisted model using the entire dataset. Figure 8 shows how in this test the LoN exhibits a leading spiral pattern, agreeing with the consensus that the LoN azimuth decreases with increasing Galactocentric radius.

Given this discrepancy between global fit and the annular fits of the LoN, it is not possible to draw any firm conclusions from the calculated LoN of the red clump sample and its implications on the warp.

4.5. Summary of Results

The fixed model has shown success in reproducing the vertical velocity trends, along with robust LoN and precession rate parameters that are consistent with recent literature (Poggio et al. 2020; Cheng et al. 2020; Jónsson & McMillan 2024). However, allowing the LoN and precession rate to vary with Galactocentric radius provides a better fit to the data in its outermost regions, and produces a warp shape that is more representative of recent observational findings (Romero-Gómez et al. 2019; Cabrera-Gadea et al. 2024). Consequently, it becomes possible to conclude that the twisted model’s additional complexity yields a more physically realistic representation of the warp found in the Milky Way, providing a description of a warp that is highly dynamic and twists across the Galactic disc. A summary of results for both models is presented in Table 1. Alongside this, a complete table of all parameters found can be seen in Appendix B.

Parameter	Value $\pm$ Uncertainty
Fixed Model	
h_{16}	4.1 ± 0.2
ϕ'_w	125.4 ± 1.0
ω_p	-12.4 ± 0.1
Twisted Model	
h_{16}	1.1 ± 0.1
$\phi'_{w,0}$	98.5 ± 1.4
$\phi'_{w,1}$	8.5 ± 0.5
$\omega_{p,0}$	-0.9 ± 0.6
$\omega_{p,1}$	-1.9 ± 0.1

TABLE 1
SUMMARY OF RESULTS FOR BOTH THE FIXED AND TWISTED MODELS.

5. DISCUSSION

5.1. Comparison to Previous Studies

Both Uppal et al. 2024 and López-Corredoira et al. 2002 use RC stars for their research and conclusively find a prominent warp in the Galactic disc (warp shapes can be seen in Figure 5). Both papers use positions rather than kinematics to model the warp and do not derive any precession rates. However, both locate a fixed LoN within 5° of the Sun’s azimuth (180°). While López-Corredoira et al. 2002 find a symmetric warp shape with amplitude of 1.23 kpc at $R = 14$ kpc, which they conclude matches that seen in the hydrogen warp, Uppal et al. 2024 reveals asymmetry within the warp using a second order model finding a Southern side maximum amplitude of -1.52 kpc and Northern side maximum

amplitude 1.30 kpc at $R = 14$ kpc.

Jónsson & McMillan 2024 used a similar modelling method to this study using a sample of roughly 27 million stars from *Gaia* DR3. They found in both their classical and classical twisted models, the warp amplitude reaches ~ 3 kpc at $R = 14$ kpc, remaining roughly constant further out. Their v_R term models, which include radial motion, do not reach any pronounced warp amplitudes, yielding a maximum vertical displacement of ~ 0.3 kpc at $R \approx 13$ kpc; however, the authors noted that these models of the warp performed significantly worse than their classical counterparts. Jónsson & McMillan 2024 also concluded a precession rate of $\sim 13 \text{ km s}^{-1} \text{ kpc}^{-1}$ in the prograde direction with a LoN around 140° , similar to the fixed model in this study.

While this study shares similar modelling with Jónsson & McMillan 2024, it benefits from significantly more robust distance measurements. Jónsson & McMillan 2024 relied on photogeometric distance measurements for a mixed stellar sample and does not account for mid-plane extinction. This contrasts with this study’s specific selection of RC stars, which, when combined with mid-infrared photometry, largely mitigates the large distance uncertainties caused by mid-plane extinction.

These results can be compared to studies utilising younger Cepheids, which trace the gas disc. Both Skowron et al. 2019 and Lemasle et al. 2022 use static spatial models to find fixed LoNs near 160° and warp amplitudes larger on the Southern side ($-0.65, -0.79$ kpc at $R = 14$ kpc respectively) compared to the Northern side ($0.54, 0.53$ kpc at $R = 14$ kpc). Cabrera-Gadea et al. 2024, also using Cepheids, use kinematics, and find a twisted LoN that varies $12.7^\circ \text{ kpc}^{-1}$ in a prograde direction, in agreement with Briggs’ rules (Briggs 1990). They also find a warp amplitude that becomes ‘almost symmetrical’ beyond 13.5 kpc, reaching a height of 0.8 kpc and a precession rate of $9.2 \pm 3.1 \text{ km s}^{-1} \text{ kpc}^{-1}$ in the prograde direction at $R = 14$ kpc. Studying the kinematics of the warp, Cabrera-Gadea et al. 2024 also discuss the LMV_z that is also twisted and offset from the LoN by a constant 25.4° , concluding that the offset is a consequence of the lopsidedness of the warp. Additionally, they find that there is a ‘plateau’ in the warp at $10 < R/\text{kpc} < 11$, a feature that has also been observed in Romero-Gómez et al. 2019; however, here it is explained as an artifact of their analytical method rather than a physical result.

These studies also agree with those using the

Milky Way’s youngest tracers, specifically hydrogen (HI) gas and OB stars. [Levine et al. 2006](#) and [Marshall et al. 2006](#) use HI gas and dust respectively, and independently find both components to be highly uneven throughout the disc, concluding that this unevenness in the disc is lopsided and a simple first-order Fourier mode is incapable of capturing the complexities of the opposing sides of the disc. [Li et al. 2023](#) look at the short-lived and massive OB stars to find a similar lopsided nature of the warp.

5.2. Model Limitations and Improvements

The models used in this study have been adopted from [Jónsson & McMillan 2024](#) with some improvements being made within this study. However, they still present limitations and possible improvements that could be made for future work, with some refinements possible to increase physical realism. First, the precession rate in the twisted model is fixed up to a certain point denoted to be where the warp begins. This fixed parameter is one that is chosen and therefore cannot vary to fit to the data. Although a warp onset radius was chosen to be 10 kpc, it may still not accurately represent the data and therefore other parameters may have to adjust for it. However, when allowing the warp onset radius to vary along with the other parameters, the base precession rate and its gradient became degenerate and were unable to conclude on a physically realistic result; therefore, the warp onset was fixed.

As well as this, the precession rate gradient was constrained to within $2 \text{ km s}^{-1} \text{ kpc}^{-2}$ beyond $R = 10 \text{ kpc}$ and throughout the modelling process the rate consistently converged at the boundaries of this constraint. When allowing for steeper changes in precession across the disc, the precession rate would simply match the new constraint. This issue is likely due to the piecewise-linear makeup of warp amplitude parameters; the flexibility this brings to the model may be too lenient and allow several mathematically possible yet physically unrealistic solutions, a notion that was problematic throughout the modelling process. Another example of this was a secondary solution to the warp precession rate, one beyond $20 \text{ km s}^{-1} \text{ kpc}^{-1}$ in the prograde direction and with it forcing unrealistic warp amplitudes. To mitigate these physically unrealistic solutions, walkers were prohibited from exploring areas in which known unrealistic solutions lay. Additional degeneracies were mitigated by forcing the warp height to be monotonically increasing outward from the

Galactic centre. Although this would not capture all the complexities of the warp as seen in higher-order models ([Cabrera-Gadea et al. 2024](#)), this adjustment from [Jónsson & McMillan 2024](#) helped break degeneracies between terms $h(R)$ and $(\bar{v}_\phi/R) - \omega_p$, increasing the realism of the results obtained.

While flexibility in the model was problematic for some parameters, rigidity in the model was also problematic for others, as seen by the discrepancy between the global twisted model LoN and the one found when the data was split into 2 kpc annuli (Figures 7 and 8). This is an indication that the global twisted model is too rigid for local orientations and therefore is forcing a trailing spiral LoN to allow for more robust parameters. This hypothesis is reinforced by the majority of research, which identifies a leading spiral LoN ([Levine et al. 2006](#); [Amôres et al. 2017](#); [Poggio et al. 2018](#); [Skowron et al. 2019](#)). It must also be noted that the fixed model LoN also does not largely agree with literature, other RC studies place the LoN near $\sim 175^\circ$ - 190° ([Wang et al. 2020](#); [Li et al. 2023](#); [Momany et al. 2006](#)) while Cepheid studies are slightly higher at $\sim 180^\circ$ - 200° ([Chen et al. 2019](#); [Huang et al. 2024](#)). It is these issues that suggest the model may not have resulted in realistic LoN parameters and therefore no conclusions can be made from the LoN obtained.

It must also be noted that a second-order Fourier model ($m = 2$) was explored to account for the lopsidedness of the warp. However, introducing this complexity consistently resulted in unrealistic expected vertical velocities in the outer disc. This model artificially inflated the warp amplitude and azimuthal gradients to fit the data, resulting in global kinematics that far exceeded known realistic velocities. Because of these unrealistic results the $m = 2$ model could not be constrained to find any realistic solutions and therefore was not included in the results for this study.

5.3. Kinematics and Shape of the Warp

Study of the vertical velocity fields shows large-scale corrugations through the disc, in both young and older stellar samples ([Poggio et al. 2025](#)), in agreement with those seen in the dataset used in this study. The improved fit suggests that by allowing the LoN to vary linearly with Galactocentric radius, the model is able to capture the advancing phase of the warp across the disc. This advance of the LoN is largely accepted to be a leading spiral ([Cheng et al. 2020](#); [Jónsson & McMillan 2024](#); [Cabrera-Gadea et al. 2024](#); [Zhou et al. 2025](#)), in agreement

with our reconstruction in Figure 8 when fitting the data split into 2 kpc annuli. This leading spiral is signature of the differential precession found by the twisted model and is indicative of a warp that is actively evolving and ‘winding’ throughout its lifetime (Poggio et al. 2020; Dehnen et al. 2023).

The complexity of the warp is further highlighted by the phase shift observed between the LMV_z and the LoN. Although the models used in this study do not account for a shift in these two lines, the shift of $\sim 25^\circ$ found by Cabrera-Gadea et al. 2024 is suggestive of a warp that is propagating through the disc. Furthermore, this finding reinforces the requirement of an $m = 2$ mode to capture the real nature of the warp, with first-order Fourier modes such as the model used in this study unable to capture this shift in the LoN and LMV_z .

Beyond the twisting signatures and propagation of the warp, the under- and overestimation of vertical velocities seen in Figure 4 highlight the lopsided nature of the warp. This lopsidedness aligns with recent second-order Fourier analysis and observations of ‘breathing modes’ (Romero-Gómez et al. 2019; Widrow et al. 2014; Cabrera-Gadea et al. 2024).

Although the warp is a global feature across the Milky Way, the nature of the inner disc appears to be largely unperturbed and remains largely flat. The stability of this region is leveraged within the modelling and can be attributed to the high surface density of stars creating a strong, self-gravitating restoring force that opposes perturbations caused by external forces (Sparke & Casertano 1988; Dubinski & Chakrabarty 2009). This is not the case in the outer regions of the Milky Way where the gravitational potential is dominated by the dark matter halo (Sofue & Rubin 2001); in these regions, external perturbations can cause large scale oscillations in the disc similar to what has been observed in the warp.

5.4. Age Dependency and Origin of the Warp

The time evolution of the warp and its dependence on the age of stellar tracer used has no clear answer. Cheng et al. 2020 find that different age populations show similar warp characteristics, differing only in velocity amplitudes. This lack of age dependency is reflected in Poggio et al. 2018 comparing upper main sequence (UMS) and giant stars and concluding that this must be a consequence of a gravitationally induced warp. For a non-gravitationally induced warp such as infalling gas or magnetic fields, it would be expected that the younger populations

who have adopted their kinematics from the HI gas exhibit more pronounced warp amplitudes.

A decrease in amplitude with older populations has been reflected in some studies (Chen et al. 2019; Chrobáková et al. 2020; Wang et al. 2020; Li et al. 2023), while interestingly, some also find that the warp is more prominent in more aged populations (Amôres et al. 2017; Romero-Gómez et al. 2019). It is these two classifications of gravitational and non-gravitational warp formation that are proposed by Skowron et al. 2019, with the gravitational mechanisms mainly including dark matter halo-disc misalignment and tidal interactions with the satellite galaxies. Han et al. 2023 argue that a dark matter halo tilted in the same direction as the stellar halo is sufficient to induce a warp in both gas and stars similar to what is observed in the Milky Way. This idea is reinforced by Han et al. 2022, who find that such tilted dark matter halos are common and can arise from a plethora of astrophysical processes.

On the other hand, Laporte et al. 2019 and Khanna et al. 2019 use simulations to find that tidal interactions from the passages of the satellite galaxy Sagittarius (Sgr) can cause excitations in the disc that reproduce signatures seen in observed data. However, this was opposed by the simulation done by Bennett et al. 2022 who use different masses of Sgr and clearly show that these interactions are not sufficient to cause the observed perturbation. To therefore supplement these individual mechanisms and help reproduce the complex Galactoseismology that we observe in the Milky Way, Weinberg & Blitz 2006 argue that the warp may have arisen from a combination of tidal forces from Sgr along with a gravitational wake induced by the passage of the satellite galaxy in the dark matter halo. Along a similar line, Li et al. 2023 conjecture that both gravitational and non-gravitational mechanisms play a role in the formation of the warp, also noting that different stellar populations exhibit different properties.

Alongside the origin of the warp, the debate over whether the warp is a transient dynamic feature or a long-lived structural component of the disc remains unsettled. Poggio et al. 2020 finds that the direction and magnitude of the warp’s precession rate favours a recent or ongoing interaction with a satellite galaxy, rather than a structural feature that has been long present in the disc. This idea is reflected in both Cheng et al. 2020 and Gómez et al. 2012 who find that the complexities in the disc have likely resulted from recent gravitational interactions

(< 4 Gyr). Opposing these ideas, there are several studies that find the warp can exist as a long-lived feature of the disc (Roškar et al. 2010; Han et al. 2023). In a similar situation to the origin of the warp, several studies find that the warp and complex morphology of the disc may be an outcome of both long-lived and transient features of the disc. Khachatryan et al. 2022 finds that long-lived interactions such as gas infall can induce sharp transient bending waves that shear and ripple through the disc while Wang et al. 2020 find that the warp is likely a long-lived feature; however, it exhibits evolving behaviours and is non-steady.

5.5. Future Work

It must be recognised that the richness of the data used in the present work allows for more detailed analysis using a more complex second order Fourier model ($m = 2$). While an initial exploration of this model was attempted, reliably constraining the parameters to find realistic results requires more extensive analysis, which serves as a natural next step for future research. The use of these more complex models would capture the asymmetries between the Northern and Southern sides of the warp and therefore allow for closer mapping of the data onto the models.

Alongside this, the use of extinction mapping in Khanna et al. 2025 at large Galactocentric radii has been limited by the distance resolution of current maps. The mid-plane dust within the Milky Way is severely obscuring the view to many stars and therefore the ability to correct for this dust is still very limited by complex extinction laws (Lallement et al. 2022). While modern dust maps are rapidly improving our view of the extinction throughout the Galactic mid-plane, future *Gaia* data releases will provide observational data that will unequivocally improve the quality and robustness of current mapping. Moreover, future *Gaia* data releases will be directly fundamental in the mapping of the kinematics and spatial geometry of the warp we see today.

6. CONCLUSIONS

This work utilises the catalogue of RC stars from Khanna et al. 2025 that comprises the combination of kinematics from *Gaia* DR3 and photometry from AllWISE to model the warp in the Milky Way. The effective spatial range of the sample was increased by calculating line-of-sight velocities of samples using the assumption of zero Galactocentric radial velocity for 3 860 894 stars that were not sufficiently bright to obtain the velocity observationally. Two

kinematic models were used to model the disc: a fixed model where the LoN and precession rate are fixed across all Galactocentric radii, and a twisted model where both of these parameters were allowed to vary linearly across the disc. Vertical velocities across the disc were calculated using Jeans' first equation of fluid continuity (Jeans 1915) assuming negligible contribution of radial velocity to vertical velocity, and these vertical velocities were imposed onto an antisymmetric, rigidly precessing disc.

It was found that the fixed model predicted steep warping in the outskirts of the galaxy reaching a maximum warp amplitude of 4.1 ± 0.2 kpc at Galactocentric radius $R = 16$ kpc. Additionally, a constant prograde precession rate of -12.4 ± 0.1 $\text{km s}^{-1} \text{kpc}^{-1}$ and a LoN of $125.4^\circ \pm 1.0^\circ$ were derived.

The twisted model found more moderate warping of the disc that plateaued beyond $R \approx 14 - 15$ kpc reaching a maximum warp amplitude of 1.1 ± 0.1 kpc at $R = 16$ kpc. The precession and LoN parameters were favoured to vary across the disc in this model, resulting in a differential precession of -0.9 ± 0.6 $\text{km s}^{-1} \text{kpc}^{-1}$ at $R = 10$ kpc that increased by 1.9 ± 0.1 $\text{km s}^{-1} \text{kpc}^{-2}$ both in the prograde direction. The resulting LoN was $98.5^\circ \pm 1.4^\circ$ at $R = 10$ kpc and formed a trailing spiral twisting $8.5^\circ \pm 0.5^\circ \text{kpc}^{-1}$. This result however was contradicted by results found when splitting the dataset into 2 kpc annuli and calculating for each annulus a fixed LoN, here finding that the set of lines formed a leading spiral in the direction of stellar rotation. From this, no clear conclusions can be made of the twisted model's LoN as the global model may be too rigid.

Comparing the two kinematic models used, it was concluded that the twisted model's allowance of a varying LoN and precession rate not only fits closer to the data but is also a more physically realistic model that better reflects the likely morphology of the disc and its twisting warp. While no clear conclusions can be made of the origin of the warp from the results of this model, they help to understand its twisted, evolving nature.

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APPENDIX A. VELOCITY ASSUMPTION

To greatly increase the effective spatial range of the dataset and therefore improve robustness of the fit of the data to the model, line-of-sight velocities were estimated using the assumption of zero Galactocentric radial velocity. This assumption allowed for the calculation of the missing velocities, and therefore made these otherwise incomplete sources viable for kinematic modelling. The method of this calculation is detailed in Section 2.1. It is important to understand the uncertainties introduced to the data when making this assumption, as it can decide whether these calculated line-of-sight velocities are improving the dataset or should not be included due to unreliability. To do this, the same assumption was made on a random selection of 2 000 000 sources from the mock dataset GeDR3Mock (Rybizki et al. 2020).

Because the exact 2D Gaussian probability cut could not be applied to the mock source selection process, a substitution was made, ensuring a 3σ cut

around the absolute magnitude theoretical centroid ($-0.16 < M_G < 1.04$, $-1.98 < M_{W1} < -1.38$), resulting in a total mock sample of 1 318 926 RC stars.

The same line-of-sight calculation process was repeated and the resulting uncertainties are shown in the top panel of Figure A1. The mean systematic uncertainties introduced to the line-of-sight velocities are generally small; however, notable exceptions occur near $l = 90^\circ$ and $l = 270^\circ$. These localised spikes are expected artifacts, occurring where the line-of-sight vector becomes nearly perpendicular to the Galactocentric radial velocity vector.

Crucially, these line-of-sight uncertainties do not propagate directly into the Cartesian velocities. This is due to the high dependence of proper-motion in vertical velocity compared to a line-of-sight velocity contribution that scales as the sine of the Galactic latitude. As demonstrated in the bottom panel of Figure A1, the assumption of zero Galactocentric radial velocity introduces minimal vertical velocity errors and therefore is beneficial to make for its resulting increase in effective spatial range of the dataset.

APPENDIX B. BEST FITTING PARAMETERS

Table B1 provides the best-fitting values and associated uncertainties from the MCMC posterior distributions for the parameters of both the fixed and twisted kinematic models. Note that the LoN parameters have not been added to 180° to show the symmetric warp closer to the Sun and are the direct results of the modelling.

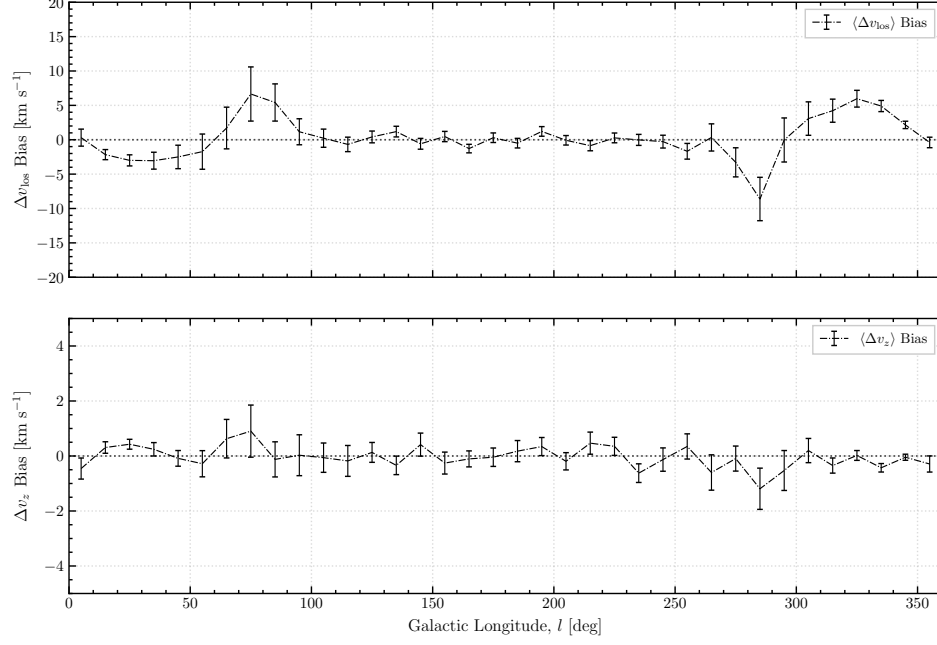


FIG. A1.— Uncertainties introduced into the line-of-sight velocity (v_{los}) when the assumption of zero Galactocentric radial velocity is made ($v_{R,gal} = 0$) for sources missing radial velocities, applied to a sampled mock catalogue of RC stars. The mean systematic bias ($\langle \Delta v \rangle$, black dash-dot line) is plotted against the left axis. Top panel: The error introduced to the estimated line-of-sight velocities (v_{los}). Bottom panel: The impact of the error on the calculated vertical velocity (v_z). Note the different y-axis scales for the v_{los} and v_z plots; the assumption introduces small uncertainties to the calculated vertical velocities preserving the macroscopic kinematic signature of the warp.

Parameter	Fixed model	Twisted model
h_7	0.005 ± 0.005	0.074 ± 0.009
h_8	0.023 ± 0.012	0.084 ± 0.008
h_9	0.069 ± 0.022	0.099 ± 0.010
h_{10}	0.170 ± 0.032	0.117 ± 0.013
h_{11}	0.380 ± 0.044	0.170 ± 0.021
h_{12}	0.782 ± 0.061	0.339 ± 0.029
h_{13}	1.372 ± 0.089	0.561 ± 0.041
h_{14}	2.377 ± 0.141	0.875 ± 0.063
h_{15}	2.895 ± 0.231	1.113 ± 0.074
h_{16}	4.117 ± 0.196	1.130 ± 0.069
ϕ'_w	-54.626 ± 1.066	—
$\phi'_{w,0}$	—	-81.561 ± 1.420
$\phi'_{w,1}$	—	8.571 ± 0.529
ω_p	-12.444 ± 0.062	—
$\omega_{p,0}$	—	-0.932 ± 0.629
$\omega_{p,1}$	—	-1.890 ± 0.098

TABLE B1
BEST FITTING PARAMETERS FOR BOTH FIXED AND TWISTED MODELS WITH RESPECTIVE UNCERTAINTIES.